VDPAM 527
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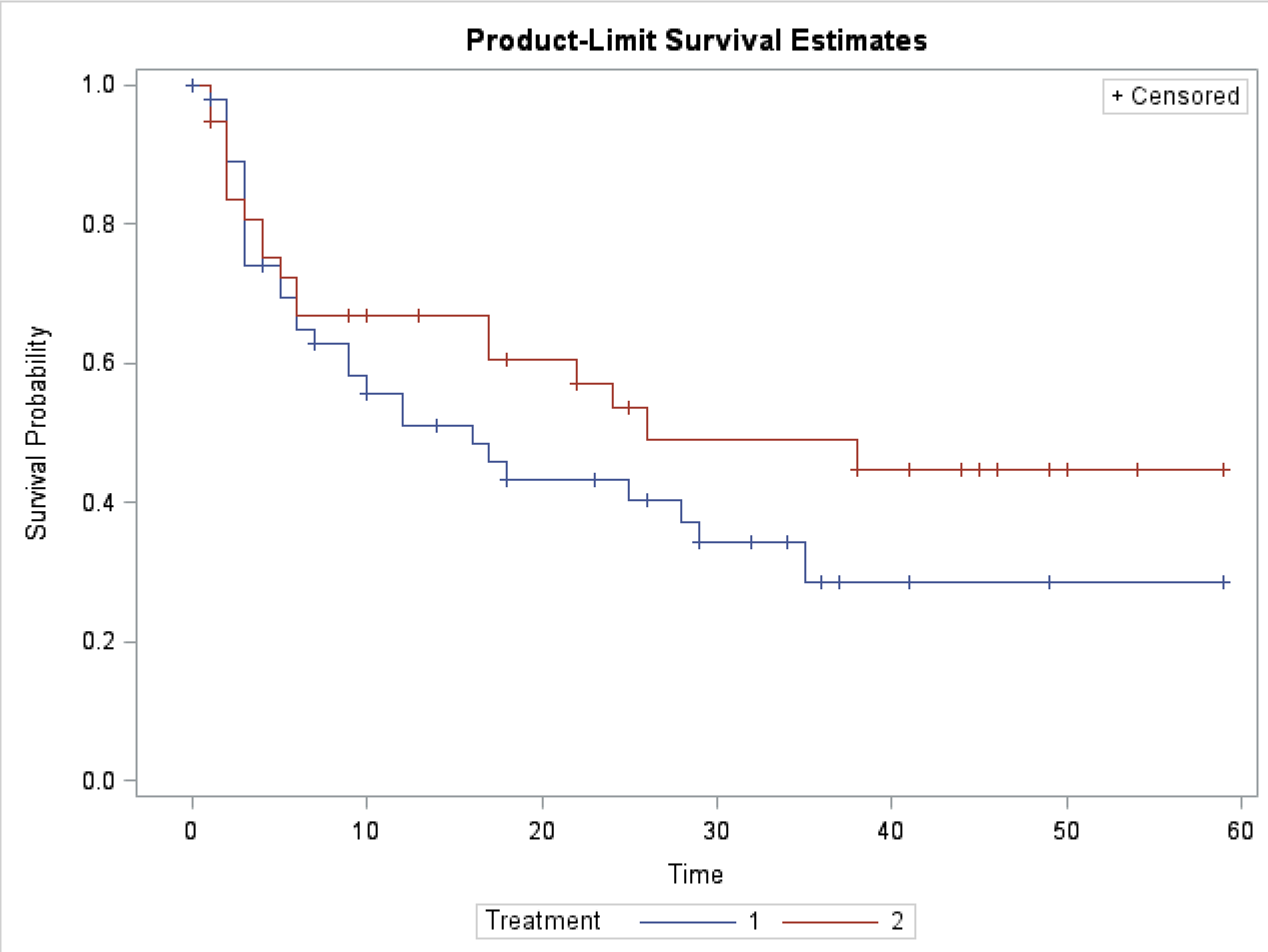
Home Work # 13
12-13-13

The data set in " http://www.public.iastate.edu/~chwang/data/Cancer.csv " comes from a paper that appeared in the Journal of the American Statistical Association by Wei et al (1989). In this study, tumors were removed from the bladders of 86 patients. Subsequently, the patients were assigned to be treated either with a placebo or with the drug thiopeta. The variables in data include:
Time: This is the time (in months) to first recurrence of a tumor or the time at which censoring took place.
Censored: This equals 1 if censoring took place and 0 otherwise.
Treatment: This equals 1 for placebo and 2 for thiopeta.
Number: This equals 1 if originally one tumor was removed and 2 if two or more tumors were removed.
You may find it useful to note the following:
Among 48 subjects receiving placebo, 29 developed new tumors while in the study.
Among 38 subjects receiving thiopeta, 18 developed new tumors while in the study.
[100] 1. Begin by examining the role of treatment without taking into account the number of tumors removed. In what follows let S1(t) denote the survival function for patients receiving placebo and S2(t) the survival function for patients receiving thiopeta.

[25] a. Create and submit a plot of the Kaplan-Meier estimates for S1(t) and S2(t). Clearly label which estimate pertains to which stage.



A + sign in both the curves indicates times when censoring is taking place.

[25] b. Interpret S1(6) and S2(6).

S1 (6) = 0.6498

This is the estimated probability (64.98%) of avoiding development of new tumor for at least full six months in the placebo group.

S2 (6) = 0.6687

This is the estimated probability (66.87 %) of avoiding development of new tumor for at least full six months in the thiopeta treated group.

[25] c. Provide point and interval estimates for S1 (6) and S2(6).

$S\ ti \pm z1-\alpha/2 \cdot se\ S\ ti$

For α=0.05 the critical value of $z_{1-\alpha/2}$=1.96
For S1(6):

0.6498 ± 1.96 (0.0707)
0.6498 ± 0.1386

Point and interval estimate S1(6)= **0.6498 (0.5112, 0.7884)**

For S2(6):

0.6687 ± 1.96 (0.0783)
0.6687 ± 0.15346

Point and interval estimate S2 (6) = **0.6687 (0.5152 , 0.8222)**

[25] d. At level 0.05 test the null hypothesis that S1(t) and S2(t) are identical against the alternative hypothesis that they differ.

H_0: S_1 (t)=S_2 (t) for every time t
H_1: S_1 (t) ≠S_2 (t) for at least one time t

Test of Equality over Strata

Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	1.5209	1	0.2175
Wilcoxon	0.7660	1	0.3815
-2Log(LR)	3.2114	1	0.0731

Chi-square= 1.5209
p-value= 0.2175

Since p-value > 0.05 we fail to reject null at significance 0.05. Therefore, we may not conclude that survival functions of placebo and thiopeta treated patients differ (at 0.05 level of significance).

SAS Output:

The SAS System

Obs	Time	Censored	Treatment	Number
1	0	1	1	1
2	1	1	1	2
3	4	1	1	1
4	7	1	1	1
5	10	1	1	1
6	6	0	1	1
7	14	1	1	1
8	18	1	1	1
9	5	0	1	2
10	12	0	1	1
11	23	1	1	2
12	10	0	1	2
13	3	0	1	1
14	3	0	1	1
15	7	0	1	2
16	3	0	1	1
17	26	1	1	2
18	1	0	1	1
19	2	0	1	2
20	25	0	1	2
21	29	1	1	2
22	29	1	1	2
23	29	1	1	1
24	28	0	1	2
25	2	0	1	2
26	3	0	1	1
27	12	0	1	2
28	32	1	1	2
29	34	1	1	1

Obs	Time	Censored	Treatment	Number
30	36	1	1	1
31	29	0	1	1
32	37	1	1	2
33	9	0	1	1
34	16	0	1	1
35	41	1	1	2
36	3	0	1	1
37	6	0	1	2
38	3	0	1	1
39	9	0	1	1
40	18	0	1	1
41	49	1	1	2
42	35	0	1	1
43	17	0	1	2
44	3	0	1	1
45	59	1	1	1
46	2	0	1	2
47	5	0	1	2
48	2	0	1	2
49	1	1	2	2
50	1	1	2	1
51	5	0	2	1
52	9	1	2	2
53	10	1	2	1
54	13	1	2	1
55	3	0	2	2
56	1	0	2	2
57	18	1	2	1
58	17	0	2	2

Obs	Time	Censored	Treatment	Number
59	2	0	2	1
60	17	0	2	1
61	22	1	2	1
62	25	1	2	2
63	25	1	2	2
64	25	1	2	1
65	6	0	2	1
66	6	0	2	1
67	2	0	2	1
68	26	0	2	2
69	38	1	2	1
70	22	0	2	1
71	4	0	2	1
72	24	0	2	1
73	41	1	2	2
74	41	1	2	1
75	1	0	2	1
76	44	1	2	1
77	2	0	2	1
78	45	1	2	2
79	2	0	2	2
80	46	1	2	2
81	49	1	2	2
82	50	1	2	1
83	4	0	2	1
84	54	1	2	2
85	38	0	2	1
86	59	1	2	2

The SAS System

The LIFETEST Procedure

Stratum 1: Treatment = 1

Product-Limit Survival Estimates

Time		Survival	Failure	Survival Standard Error	Number Failed	Number Left
0.0000		1.0000	0	0	0	48
0.0000	*	.	.	.	0	47
1.0000		0.9787	0.0213	0.0210	1	46
1.0000	*	.	.	.	1	45
2.0000		.	.	.	2	44
2.0000		.	.	.	3	43
2.0000		.	.	.	4	42
2.0000		0.8917	0.1083	0.0457	5	41
3.0000		.	.	.	6	40
3.0000		.	.	.	7	39
3.0000		.	.	.	8	38
3.0000		.	.	.	9	37
3.0000		.	.	.	10	36
3.0000		.	.	.	11	35
3.0000		0.7395	0.2605	0.0647	12	34
4.0000	*	.	.	.	12	33
5.0000		.	.	.	13	32
5.0000		0.6947	0.3053	0.0681	14	31
6.0000		.	.	.	15	30
6.0000		0.6498	0.3502	0.0707	16	29
7.0000		0.6274	0.3726	0.0717	17	28
7.0000	*	.	.	.	17	27
9.0000		.	.	.	18	26

Product-Limit Survival Estimates

Time		Survival	Failure	Survival Standard Error	Number Failed	Number Left
9.0000		0.5810	0.4190	0.0735	19	25
10.0000		0.5577	0.4423	0.0742	20	24
10.0000	*	.	.	.	20	23
12.0000		.	.	.	21	22
12.0000		0.5092	0.4908	0.0752	22	21
14.0000	*	.	.	.	22	20
16.0000		0.4838	0.5162	0.0757	23	19
17.0000		0.4583	0.5417	0.0758	24	18
18.0000		0.4328	0.5672	0.0758	25	17
18.0000	*	.	.	.	25	16
23.0000	*	.	.	.	25	15
25.0000		0.4040	0.5960	0.0760	26	14
26.0000	*	.	.	.	26	13
28.0000		0.3729	0.6271	0.0763	27	12
29.0000		0.3418	0.6582	0.0760	28	11
29.0000	*	.	.	.	28	10
29.0000	*	.	.	.	28	9
29.0000	*	.	.	.	28	8
32.0000	*	.	.	.	28	7
34.0000	*	.	.	.	28	6
35.0000		0.2849	0.7151	0.0819	29	5
36.0000	*	.	.	.	29	4
37.0000	*	.	.	.	29	3
41.0000	*	.	.	.	29	2
49.0000	*	.	.	.	29	1
59.0000	*	0.2849	0.7151	.	29	0

Note: The marked survival times are censored observations.

Summary Statistics for Time Variable Time

Quartile Estimates				
Percent	**Point Estimate**	**95% Confidence Interval**		
		Transform	**[Lower**	**Upper)**
75	.	LOGLOG	28.0000	.
50	16.0000	LOGLOG	6.0000	29.0000
25	3.0000	LOGLOG	3.0000	7.0000

Mean	Standard Error
18.2899	2.1377

Note: The mean survival time and its standard error were underestimated because the largest observation was censored and the estimation was restricted to the largest event time.

The LIFETEST Procedure

Stratum 2: Treatment = 2

Product-Limit Survival Estimates

Time		Survival	Failure	Survival Standard Error	Number Failed	Number Left
0.0000		1.0000	0	0	0	38
1.0000		.	.	.	1	37
1.0000		0.9474	0.0526	0.0362	2	36
1.0000	*	.	.	.	2	35
1.0000	*	.	.	.	2	34
2.0000		.	.	.	3	33
2.0000		.	.	.	4	32
2.0000		.	.	.	5	31
2.0000		0.8359	0.1641	0.0613	6	30
3.0000		0.8080	0.1920	0.0653	7	29
4.0000		.	.	.	8	28
4.0000		0.7523	0.2477	0.0717	9	27
5.0000		0.7245	0.2755	0.0743	10	26
6.0000		.	.	.	11	25
6.0000		0.6687	0.3313	0.0783	12	24
9.0000	*	.	.	.	12	23
10.0000	*	.	.	.	12	22
13.0000	*	.	.	.	12	21
17.0000		.	.	.	13	20
17.0000		0.6050	0.3950	0.0828	14	19
18.0000	*	.	.	.	14	18
22.0000		0.5714	0.4286	0.0848	15	17
22.0000	*	.	.	.	15	16
24.0000		0.5357	0.4643	0.0867	16	15

Product-Limit Survival Estimates						
Time		**Survival**	**Failure**	**Survival Standard Error**	**Number Failed**	**Number Left**
25.0000	*	.	.	.	16	14
25.0000	*	.	.	.	16	13
25.0000	*	.	.	.	16	12
26.0000		0.4911	0.5089	0.0902	17	11
38.0000		0.4464	0.5536	0.0924	18	10
38.0000	*	.	.	.	18	9
41.0000	*	.	.	.	18	8
41.0000	*	.	.	.	18	7
44.0000	*	.	.	.	18	6
45.0000	*	.	.	.	18	5
46.0000	*	.	.	.	18	4
49.0000	*	.	.	.	18	3
50.0000	*	.	.	.	18	2
54.0000	*	.	.	.	18	1
59.0000	*	0.4464	0.5536	.	18	0

Note: The marked survival times are censored observations.

Summary Statistics for Time Variable Time

Quartile Estimates				
Percent	**Point Estimate**	**95% Confidence Interval**		
		Transform	**[Lower**	**Upper)**
75	.	LOGLOG	.	.
50	26.0000	LOGLOG	6.0000	.
25	5.0000	LOGLOG	2.0000	17.0000

Mean	Standard Error
23.5565	2.7247

Note: The mean survival time and its standard error were underestimated because the largest observation was censored and the estimation was restricted to the largest event time.

Summary of the Number of Censored and Uncensored Values

Stratum	Treatment	Total	Failed	Censored	Percent Censored
1	**1**	48	29	19	39.58
2	**2**	38	18	20	52.63
Total		86	47	39	45.35

The SAS System

The LIFETEST Procedure

Testing Homogeneity of Survival Curves for Time over Strata

Rank Statistics

Treatment	Log-Rank	Wilcoxon
1	4.0878	174.00
2	-4.0878	-174.00

Covariance Matrix for the Log-Rank Statistics

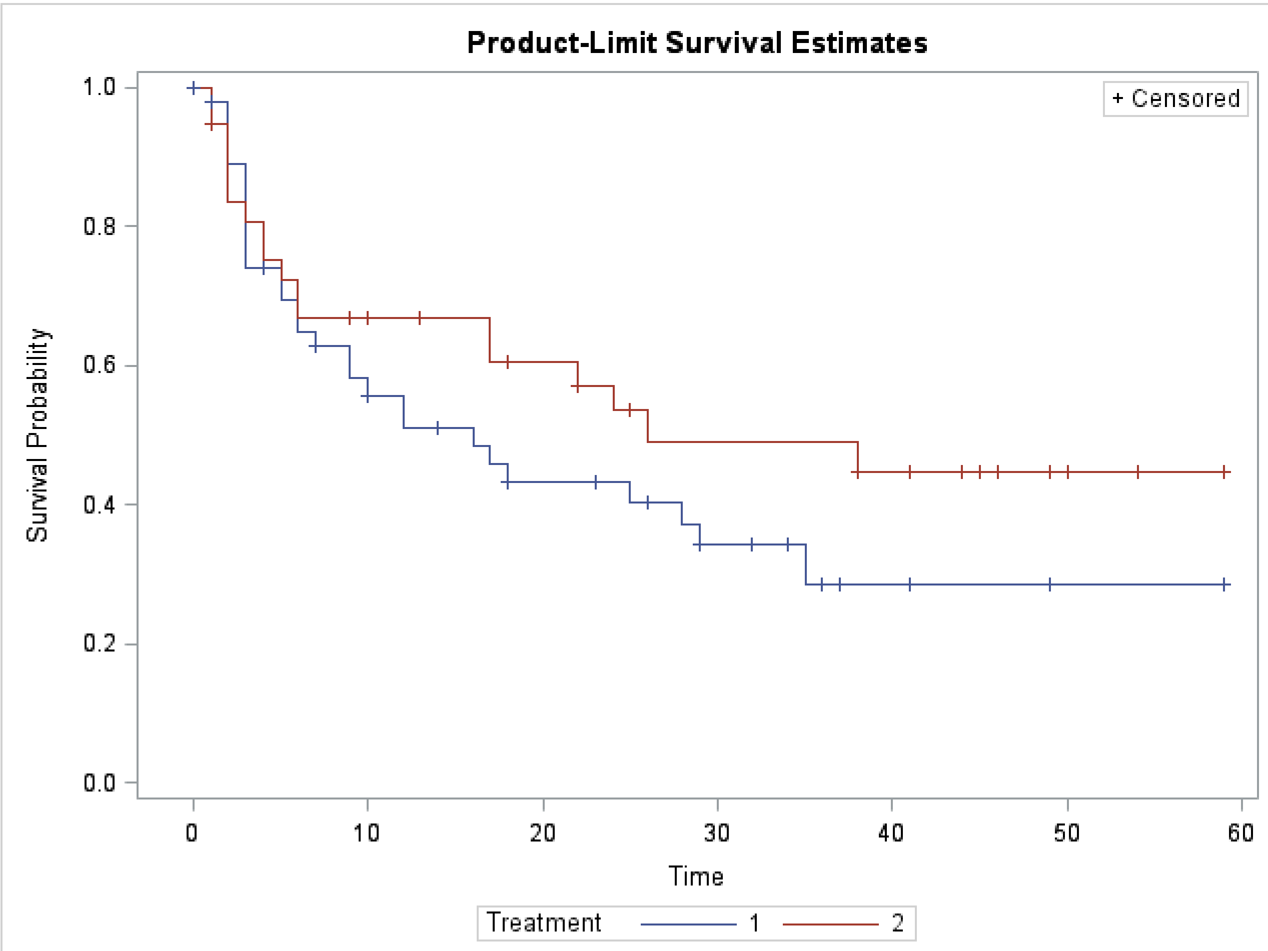
Treatment	1	2
1	10.9867	-10.9867
2	-10.9867	10.9867

Covariance Matrix for the Wilcoxon Statistics

Treatment	1	2
1	39524.5	-39524.5
2	-39524.5	39524.5

Test of Equality over Strata

Test	Chi-Square	DF	Pr > Chi-Square
Log-Rank	1.5209	1	0.2175
Wilcoxon	0.7660	1	0.3815
-2Log(LR)	3.2114	1	0.0731



SAS code:

```
filename data url' http://www.public.iastate.edu/~chwang/data/Cancer.csv ';
data CR;
infile data delimiter=',' MISSOVER FIRSTOBS=2 ;
input Time Censored Treatment Number;
run;

* Printing the data set;
PROC PRINT DATA = CR;
RUN;
```

```
* Kaplan-Meier estimation of survival curves and the log rank test;

symbol1 c=blue; symbol2 c=orange;

proc lifetest data=CR plots=(s);
     time Time*Censored(1);
     strata Treatment;
run;
```